

Mike Pozulp

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Lawrence Livermore National Laboratory
7000 East Avenue, P.O. Box 808, L-170
Livermore, California 94550
United States

Personal Information

Citizenship: U.S.

Interests

Boltzmann Transport, Radiative Transfer, Plasma Physics

Education

University of California, Berkeley	Berkeley, CA	May 2025
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PhD, Applied Science & Technology

Thesis: "A Hybrid Monte Carlo-Deterministic Second Moment Method for Thermal Radiative Transfer"

Advisors: J. L. Vujic (UCB) and T. S. Haut (LLNL)

The College of William & Mary	Williamsburg, VA	May 2015
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Bachelor of Science, *magna cum laude*

Major: Computer Science

Minor: Economics

Work Experience

Lawrence Livermore National Lab	Livermore, CA	July 2015 - Present
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Position: Staff Scientist

- Numerical method and software development for the Monte Carlo Transport Project

Presentations and Publications

Lead author indicated by *

- **"Progress Towards a Hybrid Monte Carlo-Deterministic Method for Gray Thermal Radiative Transfer with Fully-Implicit Emission-Absorption Physics"** (with T. Haut, P. Brantley, S. Olivier). Presented at *ICTT*. Aix-en-Provence, France. June 14-19, 2026.*
- **"A Hybrid Monte Carlo-Deterministic Second Moment Method with Efficient Variance Reduction"** (with T. Haut, P. Brantley, S. Olivier). *Journal of Computational Physics*, June 2026.*
- **"A Proof of the Asymptotic Variance of Path Length Estimators for Single-Collision Monte Carlo Source Iteration in the Thick Diffusion Limit"** (with T. Haut). *Journal of Computational and Theoretical Transport*, Volume 55, Number 3, 271-294, April 2026.*
- **"A Hybrid Monte Carlo-Deterministic Second Moment Method for Thermal Radiative Transfer"** PhD dissertation, University of California, Berkeley. Berkeley, California. April 2025.*
- **"A Hybrid Second Moment Method for Thermal Radiative Transfer"** (with T. Haut, P. Brantley, S. Olivier, J. Vujic). In *Proceedings of M&C 2025*. Denver, Colorado. April 2025.*
- **"Progress Porting LLNL Monte Carlo Transport Codes to the AMD Instinct MI300A APU"** (with S. Dawson, R. Bleile, P. Brantley, E. Gonzalez, M. McKinley, M. O'Brien, A. Robinson, M. Yang). In *Proceedings of M&C 2025*. Denver, Colorado. April 2025.
- **"Status of Mercury and Imp: Two Monte Carlo Transport Codes Developed Using Shared Infrastructure at Lawrence Livermore National Laboratory"** (with B. Beck, R. Bleile, P. Brantley, S. Dawson, N. Gentile, E. Gonzalez, J. Grondalski, M. Lambert, M. McKinley, M. O'Brien, R. Procassini, D. Richards, A. Robinson, S. Sepke, D. Stevens, R. Vega, M. Yang). Published in *EPJ-N* and presented at *SNA+MC 2024*. Paris, France. October 2024.*

- **“Achievement of Target Gain Larger than Unity in an Inertial Fusion Experiment”** (with O. Hurricane, A. Kritcher, A. Zylstra, D. Callahan, and many others). *Physics Review Letters* 132, 065102. February 2024.
- **“Noisy Radiation Diffusion in MFEM”** (with T. Haut, P. Brantley, J. Vujic). Presented at *MFEM Community Workshop*. Internet. October 2023.*
- **“An Implicit Monte Carlo Acceleration Scheme”** (with T. Haut, P. Brantley, J. Vujic). In *Proceedings of M&C 2023*. Niagara Falls, Canada. August 2023.*
- **“Progress Porting LLNL Monte Carlo Transport Codes to Nvidia GPUs”** (with R. Bleile, P. Brantley, S. Dawson, M. McKinley, M. O’Brien, A. Robinson, M. Yang). In *Proceedings of M&C 2023*. Niagara Falls, Canada. August 2023.*
- **“Lawson Criterion for Ignition Exceeded in an Inertial Fusion Experiment”** (with O. Hurricane, A. Kritcher, A. Zylstra, D. Callahan, and many others). *Physics Review Letters* 129, 075001. August 2022.
- **“Fast Solvers for the Finite Element Method”** (with B. Muldoon). Unpublished. May 2022.
- **“Enhancements supporting IC usage of PEM libraries on next-gen platforms”** (with D. Richards, B. Ryujin). Technical Report LLNL-TR-823775, Lawrence Livermore National Laboratory, Livermore, California. June 2021.
- **“RISC-V Code Generation Comparison”** (with Y. Miyasaka). Unpublished. May 2021.*
- **“Heterogeneity, Hyperparameters, and GPUs: Towards Useful Transport Calculations Using Neural Networks”** (with P. Brantley, T. Palmer, J. Vujic). In *Proceedings of M&C 2021*, 1252-1261. Raleigh, North Carolina. October 2021.*
- **“Extending 1D Transport Using Neural Nets to GPUs”** (with P. Brantley). Accepted for presentation at *SNA+MC 2020*. Tokyo, Japan. May 2020.*
- **“Transitioning the Scientific Software Toolchain to Clang/LLVM”** (with S. Dawson, R. Bleile, P. Brantley, M. McKinley, M. O’Brien, D. Richards). Accepted for presentation at *EuroLLVM 2020*. Paris, France. April 2020.*
- **“Status of LLNL Monte Carlo Transport Codes on Sierra GPUs”** (with M. McKinley, R. Bleile, P. Brantley, S. Dawson, M. O’Brien, D. Richards). In *Proceedings of M&C 2019*, 2160-2165. Portland, Oregon. August 2019.
- **“1D Transport Using Neural Nets, SN, and MC.”** In *Proceedings of M&C 2019*, 876-885. Portland, Oregon. August 2019.*
- **“Porting the Opacity Client Library to a CPU-GPU Cluster Using OpenMP4.5”** (with J. Kimko, R. Haque, L. Grinberg). In *Proceedings of SC17*. Denver, Colorado. November 2017.
- **“Introduction to Monte Carlo.”** Presented at *LLNL’s Computation Intern Seminar Series*, June, 2017 and *W&M Math Department Colloquium Series*, October, 2017.*
- **“LLNL Monte Carlo Transport Research Efforts for Advanced Computing Architectures”** (with P. Brantley, R. Bleile, S. Dawson, N. Gentile, M. McKinley, M. O’Brien, D. Richards, D. Stevens, J. Walsh, H. Childs). In *Proceedings of M&C 2017*. Jeju, Korea. April 2017.
- **“Optimizing Application I/O by Leveraging the Storage Hierarchy Using the Scalable Checkpoint Restart Library with a Monte Carlo Particle Transport Application on the Trinity Advanced Computing System”** (with G. Becker, P. Brantley, S. Dawson, K. Mohror, A. Moody, M. O’Brien). In *Proceedings of SC16*. Salt Lake City, Utah. November 2016.*
- **“Creating a Framework for Systematic Benchmarking of High Performance Computing Systems.”** In *Proceedings of SC14*. New Orleans, Louisiana. November 2014.*

EPJ-N is the European Physics Journal - Nuclear Sciences & Technologies

SNA+MC is the Joint International Conference on Supercomputing in Nuclear Applications + Monte Carlo.

ICTT is the International Conference on Transport Theory.

EuroLLVM is the European LLVM Developers’ Meeting.

M&C is the International Conference on Mathematics and Computational Methods applied to Nuclear Science and Engineering.

SC is the International Conference for High Performance Computing, Networking, Storage, and Analysis.

Computer Skills

- C/C++, Python, Java, R, Bash, MPI, OpenMP, CUDA, Git/Github, LLVM, Latex, PyTorch, Mathematica
- Linux, OS X, Windows, Solaris, Android, Web

Fellowships, Research Grants, and Contracts

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| • LLNL LEARN Research Funding (\$115,434) | 2020 January |
| • W&M Small Hall Makerspace Grant Recipient (\$700) | 2014 May |
| • ACM Student Research Competition Travel Award (\$500) | 2014 September |
| • Virginia Space Grant Consortium Grant Recipient (\$6,750) | 2013 June |

Honors, Prizes, and Awards

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| • LLNL Code Development Bronze Star Award | 2024 September |
| • LLNL Computer Science Spot Award | 2023 March |
| • LLNL Computational Physics Monthly Recognition Award | 2021 July |
| • LLNL Computational Physics Monthly Recognition Award | 2020 July |
| • LLNL Code Development Bronze Star Award | 2019 August |
| • LLNL Computational Physics Monthly Recognition Award | 2018 November |
| • Stanford CS148 Raytracing Project, 2nd Place | 2015 December |
| • NASA Ames Poster Contest, 1st Place | 2013 August |

Professional Development

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| • International Conference on Transport Theory | Aix-en-Provence, France | 2026 June 14-19 |
| • J34 Applied Computer Science Meeting | Livermore, California | 2026 February 9-13 |
| • NECDC 2025 | Livermore, California | 2025 October 13-17 |
| • M&C 2025 | Denver, Colorado | 2025 April 27-30 |
| • SNA + MC 2024 | Paris, France | 2024 October 20-24 |
| • MFEM Community Workshop | Internet | 2023 October 26 |
| • NECDC 2023 | Los Alamos, New Mexico | 2023 October 16-20 |
| • M&C 2023 | Niagara Falls, Canada | 2023 August 13-17 |
| • M&C 2021 | Raleigh, North Carolina | 2021 October 3-7 |
| • J34 Applied Computer Science Meeting | Livermore, California | 2020 February 24-27 |
| • LLVM Developer Meeting | San Jose, California | 2019 October 22-23 |
| • NSSC Fall Workshop | Livermore, California | 2019 October 7-9 |
| • M&C 2019 | Portland, Oregon | 2019 August 25-29 |
| • LLVM Developer Meeting | San Jose, California | 2018 October 17-18 |
| • J34 Applied Computer Science Meeting | Albuquerque, New Mexico | 2018 February 11-16 |
| • Supercomputing (SC) | Denver, Colorado | 2017 November 12-17 |
| • DoE CoE Performance Portability Meeting | Denver, Colorado | 2017 August 21-24 |
| • Supercomputing (SC) | Salt Lake City, Utah | 2016 November 13-18 |
| • DoE CoE Performance Portability Meeting | Glendale, Arizona | 2016 April 18-22 |
| • ATPESC | St. Charles, Illinois | 2016 July 31 - August 12 |
| • Supercomputing (SC) | New Orleans, Louisiana | 2014 November 16-21 |
| • Supercomputing (SC) | Denver, Colorado | 2013 November 17-22 |

Technical Coursework

University of California, Berkeley

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| • Finite Elements in Nonlinear Continua (ME 280B) | 2022 Spring |
| • Numerical Linear Algebra (MATH 221) | 2022 Spring |
| • Introduction to the Finite Element Method (ME 280A) | 2021 Fall |
| • Radiation Processes in Astronomy (PHY C207) | 2021 Fall |
| • Graduate Computer Architecture (CS 252A) | 2021 Spring |
| • Numerical Solution of Differential Equations (MATH 228B) | 2021 Spring |
| • Numerical Analysis (MATH 128A) | 2020 Fall |
| • Nuclear Reactor Theory (NE 250) | 2020 Fall |
| • Numerical Simulation in Radiation Transport (NE 255) | 2018 Fall |

University of California, Davis

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| • Network Architecture & Resource Management (EEC 273/ECS 258) | 2018 Fall |
| • Quantum Mechanics (PHY 115A) | 2017 Spring |
| • Analytical Mechanics II (PHY 105B) | 2017 Winter |
| • Analytical Mechanics I (PHY 105A) | 2016 Fall |

University of California, San Diego

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| • High Energy Density Physics (MAE 207) | 2017 Fall |
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Stanford University

- Partial Differential Equations in Engineering (CME 204)
- Compilers (CS 143)
- Introduction to Computer Graphics (CS 148)

2018 Winter
2016 Spring
2015 Fall

The College of William & Mary

- Random Walks in Biology (APSC 456)
- Reliability (CS 668)
- General Physics II, Honors (PHYS 102H)
- Analog Electronics (PHYS 252)
- Ordinary Differential Equations (MATH 302)
- General Physics I, Honors (PHYS 101H)
- Digital Electronics (PHYS 351)
- Finite Automata (CS423)
- Operating Systems (CS 424)
- Applied Financial Derivatives (ECON 415)
- Probability (MATH 401)
- Numerical Analysis (MATH 413)
- Programming Languages (CS 312)
- Systems Programming (CS 415)
- Econometrics (ECON 308)
- Multivariable Calculus (MATH 212)
- Algorithms (CS 303)
- Computer Organization (CS 304)
- Intermediate Microeconomics (ECON 303)
- Software Development (CS 301)
- Database Systems (CS 321)
- Intermediate Macroeconomics (ECON 304)
- Linear Algebra (MATH 211)
- Data Structures (CS 241)
- Discrete Structures (CS 243)

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